

TSAC: Group Project

Due Date: May 16st, 2026.

Projects turned in after 16/05/2026 at 5.00pm will receive at most 1/2 credit.

1 Introduction

Welcome to the Data Classification Challenge! This project aims to provide you with hands-on experience in time series classification techniques and methodologies.

2 Objectives

The main objectives of this project are as follows:

- To understand the principles of time series classification.
- To explore various classification algorithms and techniques.
- To apply machine learning and data mining methods for classification tasks.
- To evaluate and compare the performance of different classification models.
- To gain practical experience in working with real-world data sets.

3 Data Set

The dataset contains the power consumption of typical appliances. The classes correspond to 10 categories of home appliances: mobile phones (via chargers), coffee machines, computer stations (including monitor), fridges and freezers, Hi-Fi systems (CD players), lamp (CFL), laptops (via chargers), microwave ovens, printers, and televisions (LCD or LED).

4 Challenge Description

Your task is to build a classification model that can accurately predict the class appliance based on their power consumption. You are free to choose any classification algorithm or combination of algorithms for this task. However, you are required to justify your choice of algorithms and provide a rationale for the selected approach.

5 Evaluation Criteria

Your model will be evaluated based on its performance in terms of classification accuracy. Additionally, you may be asked to provide insights into the interpretability, scalability, and computational efficiency of your model. The evaluation will be conducted using a separate test data set, which will be provided to you without classes. So you will need to provide us with the vector containing the classification output so that we calculate the accuracy metrics.

6 Submission Guidelines

- The submission will be a notebook on Google colab documenting the details of the approach used for the classification of the data as well as all the codes developed for the task.
- Prepare a report documenting your approach, methodology, experimentation, results, and discussion (no more than 10 pages). Submit your report along with any supplementary materials (e.g., diagrams, tables, charts). Ensure that your report is well-organized, clearly written, and properly formatted.
- Include any code/scripts developed for data preprocessing, model building, and evaluation. The code needs to be clearly written with detailed comments of each part. It needs to be readily usable for the test set. The output of the code needs to be a single vector containing the predicted classification of the input data.

7 Grading Criteria

The grading for this project will be based on the following criteria:

1. Problem Understanding, Approach, Data Preprocessing, and Model Building (10 points)
2. Evaluation and Performance (4 points)
3. Analysis and Interpretation (3 points)
4. Presentation and Documentation (3 points)